José Manuel González Rodríguez

Senior Data Scientist / Machine Learning Engineer

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> Senior Data Scientist & ML Engineer (9+ years). Design, build and deploy machine learning and LLM systems end-to-end in production for enterprise clients in energy, insurance, construction and telecoms — from ETL and feature pipelines to deployed systems, evaluation frameworks and monitoring. Track record of measurable impact (5x faster document processing, 80% reduction in process automation time). ML Lead owning model KPIs and key technical decisions, working with data, platform and business stakeholders.

TECHNICAL SKILLS

Languages	Python · SQL · Bash · R · Rust (side projects)
ML / AI	Machine Learning · NLP · Computer Vision / OCR · LLMs · RAG · Prompt Engineering · LangChain · LangGraph · scikit-learn · XGBoost · Time-series Forecasting
Data Engineering	ETL / ELT · Data Pipelines · Data Quality & Validation · Pandas · PySpark · PostgreSQL / pgvector · APIs (REST / Async)
Cloud & MLOps	AWS (Bedrock, Textract, S3, SageMaker, Athena, DynamoDB, Secrets Manager) · Azure (OpenAI, Data Factory, Data Lake, Entra ID) · Docker · Kubernetes (EKS / AKS) · CI/CD (GitHub Actions) · SonarQube · Git
Observability & Quality	Elastic APM · ELK / Kibana · CloudWatch · Load Testing (Locust) · Unit & Integration Testing (pytest)
Leadership	ML Team Lead · Agile / Scrum · Stakeholder & Cross-functional Alignment · Technical Communication · Mentoring

EXPERIENCE

Machine Learning Lead

Feb 2023 – Present · Madrid, Spain

Minsait / Indra

ScottishPower / Iberdrola — 4 AI systems built and deployed

SPEN Document Library AI — *RAG/LLM assistant over 2,500+ documents (AWS Bedrock)*

Natural-language search, summarisation and Q&A over 2,500+ SPEN engineering and regulatory documents (SharePoint), refreshed on a daily (24h) cycle; hybrid retrieval (cosine similarity + Postgres full-text) over pgvector, generation on Claude Sonnet 4 (AWS Bedrock) with Nova Pro fallback, Bedrock Guardrails, LangGraph orchestration, deployed with Docker on AWS EKS across DEV/PRE/PRO.

- Machine Learning Lead; owned the ingestion subsystem end-to-end: SharePoint Microsoft Graph API client with change detection (new / updated / deleted), S3 staging, LibreOffice document conversion, PostgreSQL metadata and state tables, plus a second multimodal stage — AWS Textract layout parsing, LLM-based image description, chunking — into pgvector embeddings (Amazon Titan v2).
- Built the Ofgem reference ingestion with content-hash document versioning, so the regulatory corpus can be re-indexed idempotently.
- Established the backend test and CI harness: 250 pytest tests running in GitHub Actions with SonarQube, including the boto3/pytest mocking needed to keep the suite green in CI.
- Contributed to production hardening: Locust load testing, model fallback / retry budgets and per-question latency telemetry persisted to the database.

Ofgem ED3 Regulatory Compliance Engine — *LLM-as-judge + auto-redrafting*

- Translated a rubric defined by the business and compliance teams into a production AI review system: scores submissions against the Ofgem ED3 regulatory themes with per-theme pass/partial/fail thresholds, retrieves the applicable guidance by cosine similarity and automatically redrafts non-compliant sections, producing a redrafted DOCX/PDF plus a compliance report.
- Built the 4-phase assessment pipeline (score → gaps → replacement text → cross-section consistency) with strict JSON schemas validated by Pydantic, an async redraft stage with concurrency control on LLM calls, and Word table/section parsing into the assessment digest.
- Designed the chunked upload API for large submissions (files over 1MB split into ~900KB chunks, up to ~90MB) and instrumented each stage with perf_counter metrics for latency tracking.

Non-Conformity Email Action Tracker — *LLM microservice (Azure OpenAI)*

- Microservice that summarises offshore-wind non-conformity email threads and extracts structured action items (owner organisation, description, due date) as machine-readable JSON, consumed by an Azure Data Factory pipeline; deployed on Azure AKS with Docker and Elastic APM.
- Owned production reliability of the LLM output contract: robust JSON parsing of model responses with graceful fallbacks, structured exception handling, timeouts, token-limit tuning and adaptation to a new OpenAI API version.
- Maintained the 17-test pytest suite covering the ADF payload contract and failure modes.

ACONEX Document Compliance Pipeline — *OCR + LLM batch validation*

- Automated pre-review pipeline that pulls every new/modified engineering document from Oracle ACONEX and validates it against document-specific technical requirements before human review: AWS Textract QUERIES/LAYOUT extraction + Claude Sonnet 4.5 (Bedrock) for visual/semantic checks, a declarative rule engine (15 reusable check definitions → 12–14 checks per document type) with fuzzy/date/boolean comparators and per-check confidence → compliance score and ready-for-review flag. Batch flow: async ingestion → LibreOffice → S3 → Textract/Bedrock → PostgreSQL state machine → JSON in Azure Data Lake, orchestrated by BMC Control-M on EKS.
- Took the pipeline from functional prototype to production-grade performance and scalability, owning the processing and ingestion stages:
  - Created the memory monitoring and optimisation layer after the processing stage proved memory-bound: memory monitor utility, streaming of Textract responses, tuned Bedrock page limits, per-file size validation.
  - Built a profiling framework (cProfile and memory profiling for both ingestion and processing stages, cloud/S3 profiling, profile comparison and automated reports) to measure where time and memory were going before optimising.
  - Introduced incremental processing — delta-window filtering, download size limiters and an ingestion-progress migration — so the daily batch no longer reprocessed the whole corpus.
  - Extended the validation engine: created the content validator, extended the comparison engine, and added rejected-document checks requested by the document-control team.
  - Fixed end-to-end result delivery: Azure Data Lake upload path, JSON output persistence to S3 and processing-state consistency.

Stack: Python · FastAPI · AWS Bedrock (Claude, Nova, Titan) · Azure OpenAI · AWS Textract · LangChain · LangGraph · PostgreSQL / pgvector · Pandas · Docker · Kubernetes (EKS / AKS) · GitHub Actions · SonarQube · Elastic APM · Azure Data Factory / Data Lake · SQL · Git

Mapfre — insurance

- Machine Learning Lead in cross-selling projects for an insurance company.
- Applied state-of-the-art ML models for prediction and classification, developing end-to-end pipelines from data extraction to training and evaluation.
- Designed the model benchmarking and monitoring system: comparison of model variants and tracking of their results over time, with reporting of model performance.

Stack: Python · Pandas · Docker · XGBoost · scikit-learn · Gitflow · Tesseract · OpenCV · AWS · SQL · Git

Endesa / Enel Spain — energy

- Lead developer of 2 projects predicting energy consumption and billing for large corporate clients.
- Refactored and maintained 3 legacy projects, automating deployment and results visualisation, and proposing ML and Python best practices that improved reproducibility, portability and CI/CD.
- Team Manager of internal and external staff, more than doubling the team in less than a year while onboarding and mentoring new members.
- Technical point of contact between the development team and business stakeholders.

Stack: Python · Pandas · Random Forest · scikit-learn · Gitflow · Azure · Bitbucket · SQL · Git

Dragados — construction

- Lead of multiple cross-functional projects for automated data extraction from construction documentation, reducing documentation handling time by 5x.
- Designed and developed an ML pipeline from scratch: Computer Vision for OCR and text parsing, plus automation of the ETL process.
- Coach and advisor to the team on Agile principles and an MVP, user-centered mindset, ensuring quality and knowledge sharing — establishing new communication channels and daily knowledge sharing across the whole development team.

Stack: Python · Pandas · Docker · Tesseract · OpenCV · Azure · Bitbucket · SQL

Machine Learning Engineer & Data Scientist

Aug 2021 – Feb 2023 · Madrid, Spain

Sanz Clima

- Developed ML projects for business clients focused on prediction and optimisation, generating value for our clients.
- Extracted, cleaned, transformed and served large volumes of data using AWS cloud services (S3, Boto3, Athena, DynamoDB).
- Launched an interactive dashboard application with a consumer-centric GUI, becoming a key product for the sales department.

Stack: Python · PySpark · AWS (S3, Athena, SageMaker, DynamoDB) · scikit-learn · Docker · Linux · Flask · KNN · K-Means · Git

ML & Data Scientist

Aug 2020 – Aug 2021 · Madrid, Spain

Ayscom

- Increased data visualisations delivered to clients by 60%.
- Automated data extraction processes, reducing processing time by 80% (from minutes to seconds).
- Built NLP-based data cleansing processes, replacing a manual workflow with an automated one and improving output quality.

Stack: Python · Pandas · NLP · XML · Linux · CassandraDB · SQL · Grafana · SSH

Data Analyst

Aug 2017 – Aug 2020 · Madrid, Spain

Sanitas

- Extracted, transformed and processed company data to produce reporting, presentations and visualisations.
- Created automated data extraction and pre-processing routines and maintained the historical reporting database.

Stack: Excel · Tableau · Power BI · SQL

EDUCATION

PhD in Developmental Psychology and Machine Learning — Complutense University 2017 – 2020

Master in Data Science and Machine Learning — Complutense University 2019 – 2020